

Data Collection and Preprocessing Phase

Date	07 Oct 2025
Team ID	Xxxxxx
Project Title	Predicting Plant Growth Stages with Environmental and Management Data Using Power BI
Maximum Marks	3 Marks

Data Quality Report Template

The Data Quality Report Template will summarize data quality issues from the selected source, including severity levels and resolution plans. It will aid in systematically identifying and rectifying data discrepancies.

Data Source	Data Quality Issue	Severity	Resolution Plan
Kaggle – Plant Growth Data Classification Dataset	Missing values found in key attributes such as temperature, humidity, and sunlight hours.	High	Imputed missing numeric values using mean/median substitution and categorical ones using mode to maintain dataset balance.
Kaggle – Plant Growth Data Classification Dataset	Duplicate entries detected for certain plant IDs.	Moderate	Identified duplicates using pandas' drop_duplicates() and retained only the first unique record for each plant ID.
Kaggle – Plant Growth Data	Inconsistent data formats (e.g., numeric values stored	Moderate	Converted text-based numeric columns to float type and

Classification Dataset	as text, inconsistent date formats).		standardized date formats using Python datetime conversion.
Kaggle – Plant Growth Data Classification Dataset	Presence of outliers in temperature and sunlight hours data due to measurement errors.	Low	Detected outliers using Interquartile Range (IQR) and Winsorization techniques; replaced extreme values with acceptable thresholds.
Kaggle – Plant Growth Data Classification Dataset	Unbalanced class distribution across plant growth stages (some stages underrepresented).	Moderate	Applied resampling techniques such as SMOTE (Synthetic Minority Oversampling Technique) to ensure balanced training data.
Kaggle – Plant Growth Data Classification Dataset	Missing or unclear column names affecting readability.	Low	Renamed ambiguous column headers with descriptive names (e.g., temp_avg, humidity_percent, growth_stage).